BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 108 · Linear Search

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why must a trace table follow executed assignments in order?

Q2. What does lossy compression remove?
A. Only the file's name
B. All binary digits
C. Only duplicate file copies
D. Some original information

Q3. “Changing a variable name automatically fixes every logic error.” Correct this. Explain why.

Q4. A program is compressed before being sent. “Lossless compression removes source-code characters.” Correct this. Why does it matter here?

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Linear search — Checks one element after another.

Target — Value being searched for.

Index — Zero-based position of an array element.

Comparison — Test of current element against target.

New learning

Works without sorting: Linear search inspects items in sequence, so the list does not need to be sorted.

Check each element: Compare the target with the item at index 0, then increase the index if there is no match.

Report the result: Stop on the first match and return/report its index, or report not found after the last item.

Count comparisons: For n items, a search may check up to n items; finding an early match can require fewer.

Worked example

Trace linear search for target 14 in [6,12,9,14,20,3,11]. Use zero-based indexes. [4 marks]

1. At index 0

2. 6 ≠ 14, so continue to the next element.

3. Index 1 = 12 and index 2 = 9 are not the target.

4. Index 3 = 14; found at index 3 after four comparisons. [4]

We Do · Practise with guidance

Guided 1

Find 15 in [4,8,12,15,19]. Give its index and comparisons. [4]

Worked method: Compare 4,8,12,15 [2]; target found at index 3 [1], after four checks [1].

Guided 2

Find 7 in [9,3,8,2]. Describe the outcome. [3]

Worked method: All four values are checked [2]; target not found [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Each variable value depends on the state produced by earlier executed instructions.

Q2. D — Some original information

Q3. Clear names show what values mean and help people read and change code.

Q4. Lossless compression restores every character exactly. Losing characters could break the program. A thumbnail can lose some detail; source code cannot.

I Do / We Do pages · Guided practice

Q1. Compare 4,8,12,15 [2]; target found at index 3 [1], after four checks [1].

Q2. All four values are checked [2]; target not found [1].

Q3. It examines every element in sequence [1], does not rely on ordering [1] and stops at a match or the end [1].

You Do pages · Code First

```
array registers = ["MAR","MDR","PC","ACC"]
for i = 0 to 3
    if registers[i] == "PC" then
        print(i)
    endif
next i
```

Expected result: 2

You Do pages · Independent questions

Q1. Index 1 [1], following checks of 4 then 11 [1], total 2 comparisons [1].

Q2. Checks elements in order [1] without requiring sorting [1].

Q3. Four elements [1], because each item is compared [1] before concluding target absent [1].

Exit Ticket · Model answer

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Guided 3

Explain why a linear search works on an unsorted list. [3]

Worked method: It examines every element in sequence [1], does not rely on ordering [1] and stops at a match or the end [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the comparison that searches for the PC register.

```
array registers = ["MAR","MDR","PC","ACC"]
for i = 0 to 3
    if ... then
        print(i)
    endif
next i
```

write

Question 1

A linear search looks for 11 in [4,11,6,18,2]. State the index and number of comparisons. [3] Source: Original OCR J277-style practice question.

Question 2

Explain why linear search can find a target in an unsorted list. [2] Source: Original OCR J277-style practice question.

Question 3

A linear search fails to find 5 in [1,4,8,9]. How many elements are checked and why? [3] Source: Original OCR J277-style practice question.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Exit Ticket · Answer before checking

Binary search can safely discard half only when data is sorted.